

Proofs of the Conjectures and Questions of Kozma and Nitzan on Bernoulli Percolation

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Abstract

Kozma and Nitzan stated nine numbered conjectures and questions about connection probabilities for Bernoulli percolation on a finite weighted graph. Eight of them are proved here, five in a stronger form than the printed one. The ninth is false in the reading that allows a tie, by a four-vertex example, and it is proved when the given set of vertices has at most two elements. The strengthenings are the pre-FKG form of Conjecture 2, Conjecture 4 with the minimising vertex prescribed in advance, Conjecture 6 without either of its two printed hypotheses, a sharper right side in Question 5, and Question 9 for every minimising vertex.

1 Introduction

Let every unordered pair of vertices of a finite vertex set be open independently with a prescribed probability. Write $x \leftrightarrow y$ when the two vertices are joined by a path of open pairs. Kozma and Nitzan [KN24] asked how the probability that a vertex o reaches a vertex b compares with the probabilities that o reaches a finite set A and that the vertices of A reach b . A path from o to A and a path from a vertex of A to b should combine into a path from o to b , but the two paths are not independent, and no proof of the resulting inequality was known. Conjectures 1, 2, 3, 4 and 6 and Questions 5, 7 and 9 are proved here. Question 8 is answered negatively when the minimising vertex is not unique, and affirmatively when A has at most two elements; for three or more it is open.

Everything rests on one inequality. Order the vertices of a finite set T by the mean of an increasing function F of the open cluster. Let J_x be the event that x is the first vertex of T that the open cluster of o reaches. The inequality asserts

$$\sum_{x \in T} \mathbb{P}(J_x) \mathbb{E}[F(C_x)] \leq \mathbb{E}[F(C_o); o \leftrightarrow T] :$$

the mean of F at o , on the event that o reaches T , is at least the mean of F at the vertex through which T is first reached.

Two extensions of that inequality carry the whole note. The first conditions every expectation on the cluster missing a prescribed set Y of vertices. Applied with Y the single vertex a at which the mean of F is smallest, and applied to a subtracted form of F rather than to F , it gives Conjecture 4 with a prescribed in advance, and with that Conjectures 1, 2 and 3 and Questions 5 and 7.

The second extension replaces the single vertex o by a finite set R , and the cluster C_o by the union of the clusters of the vertices of R . For a set of two vertices that union does not see the state of the pair joining them, so the inequality survives when that pair is forced open, which is Conjecture 6. Question 9 follows from the same extension applied to the random set obtained by exposing the pairs at o .

Sections 2 and 3 fix the model, state the nine items and record the inputs. Sections 4 to 9 prove them, in the order 4, 2, 1, 3, 7, 5, 6, 9, 8, and Section 10 records the Lean formalization.

2 The Model and the Nine Numbered Statements

This section fixes the notation and reproduces the nine numbered items in the vocabulary of the paper.

Let V be a finite set. Let every unordered pair e of elements of V be open with probability $p_e \in [0, 1]$, independently over the pairs. A pair of weight zero is an absent edge, so this is percolation on a finite weighted graph with vertex set V . Write $\mathbb{P}$ and $\mathbb{E}$ for the resulting probability and expectation.

For vertices x and y , write $x \leftrightarrow y$ when there is a path of open pairs from x to y , and $x \nleftrightarrow y$ otherwise. Connection is reflexive, so $x \leftrightarrow x$ always. Write $C_x = \{y \in V : x \leftrightarrow y\}$ for the open cluster of x , and $\mathcal{C}_x$ for the set of open pairs with at least one endpoint in C_x . For a set K of pairs let $\text{span } K$ be the set of vertices lying in a pair of K , so that $C_x = \{x\} \cup \text{span } \mathcal{C}_x$.

For a set B of vertices put $C_B = \bigcup_{y \in B} C_y$, and write $\mathcal{C}_B$ for the set of open pairs with at least one endpoint in C_B . Write $\{x \leftrightarrow B\}$ for the event $\{C_x \cap B \neq \emptyset\}$ and $\{x \nleftrightarrow B\}$ for its complement. The indicator of an event is written $\mathbf{1}_A$ when the event has been given the name A , and $\mathbf{1}\{\cdot\}$ when it is written out. A function F from subsets of V to the real numbers is increasing if $F(K) \leq F(L)$ whenever $K \subseteq L$. Every set in this note is finite, and A is always a nonempty subset of V .

Kozma and Nitzan write $\mathbb{P}(0 \leftrightarrow b)$ where we write $\mathbb{P}(o \leftrightarrow b)$. They call a function f of a vertex and a configuration a monotone cluster property if $f(v, \omega)$ depends only on the open cluster of v and is increasing in it, and if $f(v, \omega) = f(w, \omega)$ whenever the pair $\{v, w\}$ is open. Their five conjectures and four questions are the following; there is no Conjecture 5 and no Question 6.

Conjecture 2.1 (Kozma and Nitzan, display (1)). Let o and b be vertices and let A be a nonempty set of vertices. Then

$$\mathbb{P}(o \leftrightarrow b) \geq \mathbb{P}(o \leftrightarrow A) \min_{x \in A} \mathbb{P}(x \leftrightarrow b).$$

The second conjecture replaces a product of two probabilities by the probability of an intersection, which the Harris inequality makes formally weaker.

Conjecture 2.2 (Kozma and Nitzan, display (2)). Let o and b be vertices and let A be a nonempty set of vertices. Then

$$\mathbb{P}(o \leftrightarrow b) \geq \min_{x \in A} \mathbb{P}(o \leftrightarrow A, x \leftrightarrow b).$$

Kozma and Nitzan call Conjecture 2.1 the post-FKG conjecture and Conjecture 2.2 the pre-FKG conjecture. They record the formally stronger form

$$\mathbb{P}(o \leftrightarrow b, o \leftrightarrow A) \geq \min_{x \in A} \mathbb{P}(o \leftrightarrow A, x \leftrightarrow b), \quad (1)$$

their display (3).

Conjecture 2.3 (Kozma and Nitzan, page 15). For every $\varepsilon > 0$ there is a $\delta > 0$ with the following property. Let A be a nonempty set of vertices of a finite weighted graph and let o and b be vertices of that graph. Assume $\mathbb{P}(o \leftrightarrow A) > 1 - \delta$, and assume $\mathbb{P}(x \leftrightarrow b) > 1 - \delta$ for every $x \in A$. Then $\mathbb{P}(o \leftrightarrow b) > 1 - \varepsilon$.

Conjecture 2.4 (Kozma and Nitzan, page 32). Let f be a monotone cluster property, let A be a nonempty set of vertices and let o be a vertex. Then

$$\mathbb{E}[f(o, \cdot); o \leftrightarrow A] \geq \min_{x \in A} \mathbb{E}[f(x, \cdot); o \leftrightarrow A].$$

For a pair e , write $\mathbb{P}^e$ and $\mathbb{E}^e$ for the probability and the expectation in the percolation measure in which the weight of e is replaced by one and all other weights are unchanged. Kozma and Nitzan write $G \downarrow e$ for the corresponding graph, which is the contraction of e as far as connection events are concerned.

Conjecture 2.5 (Kozma and Nitzan, displays (39) and (40)). Let A be a nonempty set of vertices, let b be a vertex, let $a \in A$ minimise $\mathbb{P}(x \leftrightarrow b)$ over $x \in A$, and let $e = \{v, w\}$ be a pair. Assume

$$\mathbb{P}(x \leftrightarrow b) \geq \mathbb{P}(x \leftrightarrow A) \mathbb{P}(a \leftrightarrow b) \quad \text{for } x = v \text{ and } x = w. \quad (2)$$

Then

$$\mathbb{P}^e(v \leftrightarrow b) \geq \mathbb{P}^e(v \leftrightarrow A) \mathbb{P}^e(a \leftrightarrow b). \quad (3)$$

Question 2.6 (Kozma and Nitzan, display (38)). Let A be a nonempty set of vertices and let o be a vertex. Are there numbers $c_x \geq 0$ for $x \in A$ with $\sum_{x \in A} c_x = 1$, not depending on b , such that

$$\mathbb{P}(o \leftrightarrow b) \geq \sum_{x \in A} c_x \mathbb{P}(o \leftrightarrow A, x \leftrightarrow b) \quad \text{for every vertex } b?$$

The last three questions all ask whether a specified vertex $a \in A$ satisfies the inequality of the paper's display (41),

$$\mathbb{P}(o \leftrightarrow b, o \leftrightarrow A) \geq \mathbb{P}(o \leftrightarrow A, a \leftrightarrow b). \quad (4)$$

Question 2.7 (Kozma and Nitzan, page 36). Does (4) hold when a minimises $\mathbb{P}(x \leftrightarrow b)$ over $x \in A$?

Question 2.8 (Kozma and Nitzan, page 36). Does (4) hold when a minimises $\mathbb{P}(x \leftrightarrow b, o \leftrightarrow A)$ over $x \in A$?

For a vertex o , let $\mathbb{P}_\star$ and $\mathbb{E}_\star$ be the probability and the expectation in the percolation measure in which every pair incident to o is closed and all other weights are unchanged.

Question 2.9 (Kozma and Nitzan, page 36). Let A be a nonempty set of vertices and let o and b be vertices. Does (4) hold, with all probabilities evaluated in the original measure, when a minimises $\mathbb{P}_\star(x \leftrightarrow b)$ over $x \in A$?

The minimand in Question 2.8 is the probability of the joint event that x reaches b and that o misses A . It is not the minimand of (4), which is treated by (1).

3 Conditional Positive Association and a Covariance Inequality

Two families of inequalities are taken as given. This section states them and records one construction that accompanies the second.

The first family is the conditional positive association of van den Berg, Häggström and Kahn [BHK06].

Theorem 3.1 (van den Berg, Häggström and Kahn). *Let s be a vertex, let X_1 and X_2 be sets of vertices not containing s and let g_1 and g_2 be nonnegative increasing functions of sets of pairs. Then*

$$\mathbb{E}[g_1(\mathcal{C}_s); s \leftrightarrow X_1] \mathbb{E}[g_2(\mathcal{C}_s); s \leftrightarrow X_2] \leq \mathbb{E}[g_1(\mathcal{C}_s)g_2(\mathcal{C}_s); s \leftrightarrow X_1 \cap X_2] \mathbb{P}(s \leftrightarrow X_1 \cup X_2).$$

A companion inequality of the same authors governs two clusters at once.

Theorem 3.2 (van den Berg, Häggström and Kahn). *Let S_1 and S_2 be sets of vertices, and let g_3 and g_4 be real functions of two sets of pairs, both increasing in the first argument and decreasing in the second. Then, with both functions evaluated at $(\mathcal{C}_{S_1}, \mathcal{C}_{S_2})$,*

$$\mathbb{E}[g_3; S_1 \leftrightarrow S_2] \mathbb{E}[g_4; S_1 \leftrightarrow S_2] \leq \mathbb{P}(S_1 \leftrightarrow S_2) \mathbb{E}[g_3 g_4; S_1 \leftrightarrow S_2].$$

Exchanging S_1 and S_2 shows that Theorem 3.2 also holds when g_3 and g_4 are both decreasing in the first argument and increasing in the second. Replacing g_4 by $-g_4$ shows that it reverses, to

$$\mathbb{P}(S_1 \leftrightarrow S_2) \mathbb{E}[g_3 g_4; S_1 \leftrightarrow S_2] \leq \mathbb{E}[g_3; S_1 \leftrightarrow S_2] \mathbb{E}[g_4; S_1 \leftrightarrow S_2], \quad (5)$$

when one of the two functions is increasing in the first argument and decreasing in the second while the other is decreasing in the first and increasing in the second.

Taking $X_1 = X_2$ in Theorem 3.1 gives the conditional positive association of two increasing functions of a single cluster,

$$\mathbb{E}[g_1(\mathcal{C}_s); s \leftrightarrow X] \mathbb{E}[g_2(\mathcal{C}_s); s \leftrightarrow X] \leq \mathbb{P}(s \leftrightarrow X) \mathbb{E}[g_1(\mathcal{C}_s)g_2(\mathcal{C}_s); s \leftrightarrow X]. \quad (6)$$

Both sides of (6) increase by the same amount when a constant is added to g_1 or to g_2 , and each takes finitely many values. The nonnegativity assumption may therefore be dropped in (6): monotonicity suffices. Taking $g_1 = g_2 = 1$ in Theorem 3.1 gives the Harris inequality for two decreasing events. The Harris inequality [H60] for two increasing events is used repeatedly and without further comment. The Ahlswede and Daykin four functions theorem [AD78] is used once, in Section 7.

The second family is a covariance inequality for one open cluster, conditioned on that cluster missing a prescribed set of vertices, and discounted by a finite list of auxiliary vertices.

Fix a vertex s , a set Y of vertices with $s \notin Y$, a list $z_1, \dots, z_\ell$ of distinct vertices and two further distinct vertices o and v , all the named vertices lying outside Y and being pairwise distinct. Write ν for the conditional law $\mathbb{P}(\cdot \mid s \leftrightarrow Y)$ and put

$$\pi_j(u) = \mathbb{P}(u \in C_{z_j} \mid z_j \leftrightarrow B_j), \quad \tau = \mathbb{P}(o \in C_v \mid v \leftrightarrow B_{\ell+1}), \quad (7)$$

where $B_1 = \{s\} \cup Y$ and $B_{j+1} = B_j \cup \{z_j\}$ for $1 \leq j \leq \ell$. For a real function q on V put $q^0 = q$ and

$$q^j(u) = q^{j-1}(u) - \pi_j(u) q^{j-1}(z_j) \quad \text{for } 1 \leq j \leq \ell, \quad \mathfrak{M}[q] = q^\ell(o) - \tau q^\ell(v). \quad (8)$$

The j th step removes from the value at every vertex the part explained by z_j . All the constants are computed under $\mathbb{P}$ and not under ν .

Theorem 3.3. *Let V be a finite vertex set with every pair weight strictly between zero and one, let $Y \subseteq V$, let s be a vertex outside Y , let $z_1, \dots, z_\ell$ be distinct vertices outside $Y \cup \{s\}$, let o and v be two further distinct vertices outside $Y \cup \{s, z_1, \dots, z_\ell\}$, and let g be an increasing function of sets of pairs. Write ν for the conditional law $\mathbb{P}(\cdot \mid s \leftrightarrow Y)$ and let $\mathfrak{M}$ be the functional (8). Then*

$$\mathfrak{M}[u \mapsto \text{Cov}_\nu(g(\mathcal{C}_s), \mathbf{1}\{u \in C_s\})] \geq 0.$$

For $\ell = 0$ the conclusion reads

$$\text{Cov}_\nu(g(\mathcal{C}_s), \mathbf{1}\{o \in C_s\}) \geq \mathbb{P}(o \in C_v \mid v \leftrightarrow \{s\} \cup Y) \text{Cov}_\nu(g(\mathcal{C}_s), \mathbf{1}\{v \in C_s\}),$$

and if in addition $Y = \emptyset$ and $g = \mathbf{1}\{v \in \cdot\}$ it is the Harris inequality.

One construction that accompanies Theorem 3.3 is used in Section 7. Condition on $\mathcal{C}_Y$. On the event $D = \{s \leftrightarrow Y\}$ the vertex set $C_Y = Y \cup \text{span } \mathcal{C}_Y$ misses C_s , every pair with exactly one endpoint in C_Y is closed, and the pairs with both endpoints outside C_Y retain their original weights and are independent of the pairs meeting C_Y .

Write $\bar{g}$ for the mean of $g(\mathcal{C}_s)$ in a configuration in which every pair meeting C_Y is closed and every remaining pair keeps its original weight. Then the random variable

$$\Theta = \mathbf{1}_D(g(\mathcal{C}_s) - \bar{g}) \quad (9)$$

has $\mathbb{E}[\Theta \mid \mathcal{C}_Y] = 0$. For a set K of vertices disjoint from $\{s\} \cup Y$, let $\Psi(K)$ be minus the mean of Θ in the configuration in which every pair meeting K is closed. Then $\Psi(\emptyset) = 0$, the function Ψ is nonnegative and increasing, and for every vertex z and every set B containing $\{s\} \cup Y$ but not z ,

$$\mathbb{E}[\Theta; z \leftrightarrow B] = -\mathbb{E}[\Psi(C_z); z \leftrightarrow B]. \quad (10)$$

Both facts express the domain Markov property of percolation: given $\mathcal{C}_B$, the pairs with both endpoints outside C_B retain their original weights and are independent of that data.

Two properties of the conditioned covariance are recorded for the same reason. For a set W of vertices write $\mathbb{P}_W$, $\mathbb{E}_W$ and Cov_W for the probability, the expectation and the covariance in the percolation measure in which every pair with an endpoint outside W is closed and every pair with both endpoints in W keeps its weight. For a set X of vertices containing s and a vertex $v \notin X$, put

$$\eta_v(W) = \text{Cov}_W(g(\mathcal{C}_s), \mathbf{1}\{v \leftrightarrow X\}) \quad \text{for } v \in W, \quad \eta_v(W) = 0 \quad \text{for } v \notin W. \quad (11)$$

Then for every W and every $Q \subseteq W$,

$$\mathbb{E}_W[\mathbf{1}\{s \leftrightarrow Q\} \eta_v(W \setminus C_Q)] \mathbb{P}_W(v \leftrightarrow X) \leq \mathbb{P}_W(v \leftrightarrow X \cup Q) \eta_v(W), \quad (12)$$

and the left side of (12) decreases as Q grows.

4 A Lower Bound Conditioned on Missing a Fixed Set

This section proves the inequality on which everything else rests.

Throughout this section V is a finite vertex set with pair weights in $[0, 1]$, the set Y is a subset of V , and F is an increasing function of subsets of V . For a vertex x with $\mathbb{P}(x \leftrightarrow Y) > 0$ put

$$m_x = \frac{\mathbb{E}[F(C_x); x \leftrightarrow Y]}{\mathbb{P}(x \leftrightarrow Y)}, \quad (13)$$

the mean of F at x given that x misses Y . Also fix a set $T \subseteq V \setminus Y$ with $\mathbb{P}(x \leftrightarrow Y) > 0$ for every $x \in T$, and an injective real function r on T with $m_x \leq m_y$ whenever $r(x) < r(y)$. Such an r exists: order T by the values m_x and break ties arbitrarily. For $x \in T$ and a vertex o write

$$J_x = \{o \leftrightarrow Y\} \cap \{o \leftrightarrow x\} \cap \bigcap_{y \in T, r(y) < r(x)} \{o \leftrightarrow y\} \quad (14)$$

for the event that o misses Y and that x is the vertex of smallest value of r that the cluster of o reaches in T . The events J_x are disjoint and their union is $\{o \leftrightarrow Y\} \cap \{o \leftrightarrow T\}$. Put

$$\Lambda_o(T) = \mathbb{E}[F(C_o); o \leftrightarrow Y, o \leftrightarrow T] - \sum_{x \in T} \mathbb{P}(J_x) m_x. \quad (15)$$

On J_x the vertex x has the smallest value of r among the vertices of T in C_o , hence the smallest value of m , and there the clusters of o and x coincide. Consequently (15) admits the expression

$$\Lambda_o(T) = \mathbb{E}\left[\mathbf{1}\{o \leftrightarrow Y, o \leftrightarrow T\} \left(F(C_o) - \min_{x \in T \cap C_o} m_x\right)\right], \quad (16)$$

which does not refer to r . In (16), and in its analogue for a set of vertices in Section 7, the minimum is declared to be zero when the set over which it is taken is empty; the value is immaterial there because the accompanying indicator vanishes.

Two positions of o are settled at once. If $o \in Y$, then the event $\{o \leftrightarrow Y\}$ is empty and $\Lambda_o(T) = 0$. If $o \in T$, then $o \in T \cap C_o$ on $\{o \leftrightarrow Y\}$, so the minimum in (16) is at most m_o , whence $\Lambda_o(T) \geq \mathbb{E}[F(C_o) - m_o; o \leftrightarrow Y] = 0$ by (13).

Theorem 4.1. *With V, Y, F, T and r as fixed above, let $o \in V$. Then $\Lambda_o(T) \geq 0$, that is*

$$\sum_{x \in T} \mathbb{P}(J_x) m_x \leq \mathbb{E}[F(C_o); o \leftrightarrow Y, o \leftrightarrow T].$$

For $Y = \emptyset$ this is the inequality quoted in the introduction. No sign hypothesis on F is needed. The proof removes from T the vertex at which m is largest, and transfers the resulting deficit back to o by the following inequality, which is what Theorem 3.3 supplies.

Proposition 4.2. *With V, Y, F, T and r as fixed above, let $v \in V$ lie outside $Y \cup T$. Then, for every vertex o ,*

$$\mathbb{P}(v \leftrightarrow Y \cup T, o \leftrightarrow v) \Lambda_v(T) \leq \mathbb{P}(v \leftrightarrow Y \cup T) \Lambda_o(T).$$

Proof. We proceed in three steps: an induction on the cardinality of T , its specialisation to an empty list of auxiliary vertices, and the passage to weights in $[0, 1]$.

Assume first that every pair weight lies strictly between zero and one, that o lies outside $Y \cup T$ and that $o \neq v$. Let $z_1, \dots, z_\ell$ be distinct vertices outside $Y \cup T \cup \{o, v\}$. Let $\mathfrak{M}$ be the functional (8) built

from the constants (7) with Y replaced by $Y \cup T$ at every step. We start by showing, by induction on the cardinality of T and simultaneously for every such list and every pair of vertices o and v , that

$$\mathfrak{M}[u \mapsto \Lambda_u(T)] \geq 0. \quad (17)$$

In the base case $T = \emptyset$ each $\Lambda_u(\emptyset)$ vanishes. For the inductive step, let T be nonempty, let $k \in T$ have the largest value of r , and put $T_0 = T \setminus \{k\}$. Abbreviate

$$\begin{aligned} P &= \mathbb{P}(k \leftrightarrow Y \cup T_0), & P(u) &= \mathbb{P}(k \leftrightarrow Y \cup T_0, u \leftrightarrow k), \\ I &= \mathbb{E}[F(C_k); k \leftrightarrow Y \cup T_0], & I(u) &= \mathbb{E}[F(C_k); k \leftrightarrow Y \cup T_0, u \leftrightarrow k]. \end{aligned}$$

Since k has the largest value of r , adding k to T_0 leaves the events J_x for $x \in T_0$ unchanged and creates the single new event $\{k \leftrightarrow Y \cup T_0, u \leftrightarrow k\}$: on $\{u \leftrightarrow k\}$ the clusters of u and k coincide. On that event $F(C_u) = F(C_k)$, so

$$\Lambda_u(T) = \Lambda_u(T_0) + I(u) - P(u) m_k. \quad (18)$$

Write $\text{Cov}^*(u)$ for $P I(u) - I P(u)$. This is the conditioned covariance of Theorem 3.3 without denominators, for the vertex k , the set $Y \cup T_0$ and the increasing function $\mathcal{C} \mapsto F(\{k\} \cup \text{span } \mathcal{C})$. Put $\gamma_k = m_k P - I$. Expanding both sides gives

$$P(I(u) - P(u) m_k) = \text{Cov}^*(u) - \gamma_k P(u). \quad (19)$$

Theorem 3.3 for the vertex k , the set $Y \cup T_0$, the same list and the same two vertices gives $\mathfrak{M}[\text{Cov}^*] \geq 0$. The same theorem for the increasing function $\mathcal{C} \mapsto \mathbf{1}\{\mathcal{C} \neq \emptyset\}$ gives $\mathfrak{M}[P] \geq 0$. Indeed $\mathfrak{M}$ evaluates its argument only at o , at v and at the vertices of the list, all of which differ from k . On $\{u \leftrightarrow k\}$ with $u \neq k$ the cluster of k contains an open pair, and on $\{k \leftrightarrow Y \cup T_0\}$ the mean of that indicator is P minus the probability that k misses $Y \cup T_0$ and lies in no open pair. The covariance in question is therefore the latter probability times $P(u)$, and that probability is positive because all weights are strictly below one.

Finally,

$$\gamma_k \leq \Lambda_k(T_0). \quad (20)$$

Indeed (13) gives $\mathbb{E}[(F(C_k) - m_k); k \leftrightarrow Y] = 0$. Splitting $\{k \leftrightarrow Y\}$ into $\{k \leftrightarrow Y \cup T_0\}$ and $\{k \leftrightarrow Y\} \cap \{k \leftrightarrow T_0\}$ turns this into $\gamma_k = \mathbb{E}[(F(C_k) - m_k); k \leftrightarrow Y, k \leftrightarrow T_0]$. Expanding the latter over the disjoint events J_x of (14) taken with k in place of o gives $\gamma_k = \Lambda_k(T_0) + \sum_{x \in T_0} \mathbb{P}(J_x)(m_x - m_k)$, in which every summand is at most zero because $r(x) < r(k)$.

Now apply $\mathfrak{M}$ to (18), multiply by $P > 0$ and use (19):

$$P \mathfrak{M}[\Lambda_u(T)] = P \mathfrak{M}[\Lambda_u(T_0)] + \mathfrak{M}[\text{Cov}^*] - \gamma_k \mathfrak{M}[P].$$

The middle term is nonnegative. Replace γ_k by the larger $\Lambda_k(T_0)$, which is legitimate because $\mathfrak{M}[P] \geq 0$, and write $P(u) = P \pi_k(u)$ with $\pi_k(u) = \mathbb{P}(u \in C_k \mid k \leftrightarrow Y \cup T_0)$. The right side is then at least P times the value of $\mathfrak{M}$ at $u \mapsto \Lambda_u(T_0)$ for the list obtained by inserting k in front of $z_1, \dots, z_\ell$, by the recursion in (8) and linearity. That value is nonnegative by the induction hypothesis. Dividing by P gives (17).

We now take $\ell = 0$ in (17). The conclusion is $\Lambda_o(T) \geq \tau \Lambda_v(T)$ with $\tau = \mathbb{P}(o \in C_v \mid v \leftrightarrow Y \cup T)$, which is the assertion after multiplication by $\mathbb{P}(v \leftrightarrow Y \cup T) > 0$. If $o = v$, then the two sides of the assertion are equal. If o lies in Y or in T , then the event $\{v \leftrightarrow Y \cup T\} \cap \{o \leftrightarrow v\}$ is empty, so the left side vanishes, while the right side is nonnegative by the two cases settled after (16).

It remains to pass to weights in $[0, 1]$. The right side of (16) is a finite sum, over the configurations, of the probability of the configuration, a polynomial in the pair weights, times a value depending on the weights only through the finitely many numbers m_x . Each of those is a ratio of polynomials whose denominator is positive at the given weight vector. Both sides of the assertion are therefore continuous at that vector. Approximate it by weight vectors with all coordinates strictly inside the unit interval and choose for each an injective r ordered by its own numbers m_x . By (16) neither side depends on that choice, so the case already proved applies to each of them, and continuity gives the inequality at the given vector. $\square$

Proof of Theorem 4.1. We prove by induction on the cardinality of T that $\Lambda_o(T) \geq 0$ for every vertex o . In the base case $T = \emptyset$ the quantity (15) vanishes. For the inductive step, let T be nonempty, let $k \in T$ have the largest value of r , put $T_0 = T \setminus \{k\}$, and keep the abbreviations P , $P(u)$, I and $I(u)$ of the proof of Proposition 4.2. The identity (18) with $u = o$ reads $\Lambda_o(T) = \Lambda_o(T_0) + I(o) - m_k P(o)$.

Apply (6) with $s = k$, with $X = Y \cup T_0$ and with the two increasing functions $\mathcal{C} \mapsto F(\{k\} \cup \text{span } \mathcal{C})$ and the indicator of the event that o lies in the cluster of k . The result is $P(o)I \leq P I(o)$, which rearranges to

$$P(m_k P(o) - I(o)) \leq P(o)(m_k P - I) = P(o) \gamma_k.$$

By (20) and $P(o) \geq 0$ the right side is at most $P(o)\Lambda_k(T_0)$, and by Proposition 4.2 with $v = k$ that is at most $P\Lambda_o(T_0)$. Hence $P\Lambda_o(T) \geq 0$. If $P > 0$, then this is the assertion. If $P = 0$, then $P(o) = 0$ and $I(o) = 0$, so $\Lambda_o(T) = \Lambda_o(T_0)$, which is nonnegative by the induction hypothesis. $\square$

5 A Prescribed Vertex Works in Conjecture 4

This section turns Theorem 4.1 into a statement in which the minimising vertex is fixed before the event $\{o \leftrightarrow A\}$ is imposed, and then reads off Conjectures 1 to 4 and Question 7.

The obstacle is that the numbers m_x of (13) are conditional means, whereas Conjecture 2.4 compares unconditional ones. The following function removes the difference. For a vertex a , an increasing F and a set K of vertices, write $\Phi_a(K)$ for $F(K)$ minus the mean of $F(C_a)$ in a configuration in which every pair meeting K is closed and every remaining pair keeps its original weight.

Lemma 5.1. *Let a and o be vertices and let F be an increasing function of subsets of V . Then Φ_a is increasing, and:*

(i) *for every family $\mathcal{S}$ of sets of pairs,*

$$\mathbb{E}[F(C_o) - F(C_a); o \nleftrightarrow a, \mathcal{C}_o \in \mathcal{S}] = \mathbb{E}[\Phi_a(C_o); o \nleftrightarrow a, \mathcal{C}_o \in \mathcal{S}];$$

(ii) $\mathbb{E}[\Phi_a(C_o); o \nleftrightarrow a] = \mathbb{E}[F(C_o)] - \mathbb{E}[F(C_a)]$.

Proof. Enlarging K increases $F(K)$ and closes more pairs, which decreases the subtracted mean, so Φ_a is increasing.

For (i), condition on $\mathcal{C}_o$. On $\{o \nleftrightarrow a\}$ the vertex a lies outside C_o , every pair with exactly one endpoint in C_o is closed, and the pairs with both endpoints outside C_o retain their original weights. Hence the conditional mean of $F(C_a)$ given $\mathcal{C}_o$ is $F(C_o) - \Phi_a(C_o)$. Both events in the statement are determined by $\mathcal{C}_o$, so summing over the values of $\mathcal{C}_o$ in $\mathcal{S}$ gives (i).

For (ii), take $\mathcal{S}$ to consist of all sets of pairs. On $\{o \leftrightarrow a\}$ the two clusters coincide, so $F(C_o) - F(C_a)$ vanishes there and the left side of (i) equals $\mathbb{E}[F(C_o) - F(C_a)]$. $\square$

Theorem 5.2. *Let A be a nonempty finite set of vertices, let $a \in A$, let o be a vertex, and let F be an increasing function of subsets of V with $\mathbb{E}[F(C_a)] \leq \mathbb{E}[F(C_x)]$ for every $x \in A$. Then*

$$\mathbb{E}[F(C_a); o \leftrightarrow A] \leq \mathbb{E}[F(C_o); o \leftrightarrow A].$$

Proof. Let T be the set of $x \in A \setminus \{a\}$ with $\mathbb{P}(x \leftrightarrow a) > 0$. Apply Theorem 4.1 with $Y = \{a\}$, with the set T , with the increasing function Φ_a and with the vertex o . By Lemma 5.1(ii) applied at x in place of o , the number (13) attached to $x \in T$ is $(\mathbb{E}[F(C_x)] - \mathbb{E}[F(C_a)])/\mathbb{P}(x \leftrightarrow a) \geq 0$. The sum subtracted in (15) is therefore nonnegative, and Theorem 4.1 gives $\mathbb{E}[\Phi_a(C_o); o \nleftrightarrow a, o \leftrightarrow T] \geq 0$. The event $\{o \leftrightarrow T\}$ is determined by $\mathcal{C}_o$, so Lemma 5.1(i) converts this into

$$\mathbb{E}[F(C_o) - F(C_a); o \nleftrightarrow a, o \leftrightarrow T] \geq 0. \quad (21)$$

Split $\{o \leftrightarrow A\}$ into $\{o \leftrightarrow a\}$, on which the integrand $F(C_o) - F(C_a)$ vanishes, and $\{o \nleftrightarrow a\} \cap \{o \leftrightarrow A \setminus \{a\}\}$. The latter differs from $\{o \nleftrightarrow a\} \cap \{o \leftrightarrow T\}$ only by configurations in which o misses a and

reaches at least one vertex x of $A \setminus \{a\}$ outside T . Such a configuration lies in the null event $\{x \leftrightarrow a\}$, and a finite union of null events is null. Hence the left side of (21) equals $\mathbb{E}[F(C_o) - F(C_a); o \leftrightarrow A]$. $\square$

Conjecture 2.4 follows by choosing a to realise the minimum of $\mathbb{E}[F(C_x)]$ over $x \in A$, provided every monotone cluster property is of the form $\omega \mapsto F(C_v(\omega))$ for an increasing F . That is the next lemma, whose hypotheses omit the requirement that $f(v, \omega)$ depend only on C_v , since that follows from the requirement that $f(v, \omega)$ be increasing in C_v .

Lemma 5.3. *Let f be a function of a vertex and a configuration such that $f(v, \omega) \leq f(v, \xi)$ whenever $C_v(\omega) \subseteq C_v(\xi)$, and such that $f(v, \omega) = f(w, \omega)$ whenever the pair $\{v, w\}$ is open in ω . Then there is an increasing function F of vertex sets with $f(v, \omega) = F(C_v(\omega))$ for every vertex v and every configuration ω .*

Proof. For a nonempty vertex set S let ω_S be the configuration in which the open pairs are exactly those with both endpoints in S . Every vertex of S has open cluster S in ω_S , and for distinct $v, w \in S$ the pair $\{v, w\}$ is open in ω_S . Hence $f(v, \omega_S)$ does not depend on the choice of $v \in S$; call the common value $F(S)$.

If $S \subseteq S_1$ are nonempty and $v \in S$, then $C_v(\omega_S) = S \subseteq S_1 = C_v(\omega_{S_1})$, so $F(S) \leq F(S_1)$. Extend F to the empty set by a value at most $F(S)$ for every nonempty S . Finally, for a vertex v and a configuration ω put $S = C_v(\omega)$; then $C_v(\omega_S) = S = C_v(\omega)$, and the hypothesis applied in both directions gives $f(v, \omega) = f(v, \omega_S) = F(S)$. $\square$

Corollary 5.4. *Conjecture 2.4 holds. In particular it holds for $f(v, \omega) = |C_v|$ and for $f(v, \omega) = \mathbf{1}\{|C_v| \geq n\}$ for every positive integer n .*

Proof. By Lemma 5.3 write $f(v, \cdot) = F(C_v)$ with F increasing, choose $a \in A$ realising the minimum of $\mathbb{E}[F(C_x)]$ over $x \in A$, and apply Theorem 5.2. The two displayed examples are increasing functions of the vertex set. $\square$

The remaining conjectures follow by specialising F to an indicator.

Corollary 5.5. *The inequality (1) holds, and so do Conjectures 2.1, 2.2 and 2.3; Conjecture 2.3 holds with $\delta = \varepsilon/2$. Question 2.7 has an affirmative answer.*

Proof. Take $F = \mathbf{1}\{b \in \cdot\}$, so that $\mathbb{E}[F(C_x); o \leftrightarrow A] = \mathbb{P}(x \leftrightarrow b, o \leftrightarrow A)$ and $\mathbb{E}[F(C_x)] = \mathbb{P}(x \leftrightarrow b)$. Corollary 5.4 is then (1), and dropping the event $\{o \leftrightarrow A\}$ on the left gives Conjecture 2.2. If a minimises $\mathbb{P}(x \leftrightarrow b)$ over $x \in A$, then Theorem 5.2 gives $\mathbb{P}(a \leftrightarrow b, o \leftrightarrow A) \leq \mathbb{P}(o \leftrightarrow b, o \leftrightarrow A)$, which is (4); so Question 2.7 has an affirmative answer.

For Conjecture 2.1, the events $\{x \leftrightarrow b\}$ and $\{o \leftrightarrow A\}$ are both increasing, so the Harris inequality gives $\mathbb{P}(o \leftrightarrow A)\mathbb{P}(x \leftrightarrow b) \leq \mathbb{P}(x \leftrightarrow b, o \leftrightarrow A)$ for every $x \in A$. Taking the minimum over $x \in A$ and using (1),

$$\mathbb{P}(o \leftrightarrow A) \min_{x \in A} \mathbb{P}(x \leftrightarrow b) \leq \min_{x \in A} \mathbb{P}(x \leftrightarrow b, o \leftrightarrow A) \leq \mathbb{P}(o \leftrightarrow b, o \leftrightarrow A) \leq \mathbb{P}(o \leftrightarrow b).$$

For Conjecture 2.3 take $\delta = \varepsilon/2$. If $\varepsilon \geq 2$, then $1 - \varepsilon \leq -1 < \mathbb{P}(o \leftrightarrow b)$ and there is nothing to prove. Otherwise $1 - \varepsilon/2$ is positive, and Conjecture 2.1 gives $\mathbb{P}(o \leftrightarrow b) > (1 - \varepsilon/2)^2 = 1 - \varepsilon + \varepsilon^2/4 \geq 1 - \varepsilon$. $\square$

6 One Coefficient Vector for Every Target

Question 2.6 asks for one probability vector on A that works for every vertex b at once, whereas Corollary 5.4 produces one vertex of A for each b . The passage from one to the other is finite-dimensional convexity. The first step applies Corollary 5.4 not to a single indicator but to a nonnegative weighted count of the vertices of the cluster.

Lemma 6.1. *Let A be a nonempty finite set of vertices, let o be a vertex and let $y_b \geq 0$ for $b \in V$. Then there is a vertex $a \in A$ with*

$$\sum_{b \in V} y_b \mathbb{P}(o \leftrightarrow A, a \leftrightarrow b) \leq \sum_{b \in V} y_b \mathbb{P}(o \leftrightarrow b, o \leftrightarrow A).$$

Proof. Put $F(K) = \sum_{b \in V} y_b \mathbf{1}\{b \in K\}$, an increasing function of vertex sets because each y_b is nonnegative. For every vertex x ,

$$\mathbb{E}[F(C_x); o \leftrightarrow A] = \sum_{b \in V} y_b \mathbb{P}(x \leftrightarrow b, o \leftrightarrow A).$$

Let $a \in A$ realise the minimum of $\mathbb{E}[F(C_x); o \leftrightarrow A]$ over $x \in A$. By Corollary 5.4 the left side of the assertion is $\mathbb{E}[F(C_a); o \leftrightarrow A]$, which is at most $\mathbb{E}[F(C_o); o \leftrightarrow A]$, and the latter is the right side. $\square$

A separation argument turns that family of vertices into a single probability vector.

Theorem 6.2. *For every finite weighted graph, every nonempty finite $A \subseteq V$ and every vertex o there are numbers $c_x \geq 0$ for $x \in A$ with $\sum_{x \in A} c_x = 1$ such that*

$$\sum_{x \in A} c_x \mathbb{P}(o \leftrightarrow A, x \leftrightarrow b) \leq \mathbb{P}(o \leftrightarrow b, o \leftrightarrow A) \leq \mathbb{P}(o \leftrightarrow b) \quad \text{for every vertex } b.$$

In particular Question 2.6 has an affirmative answer.

Proof. The second inequality is event inclusion, so only the first has to be proved. For each $x \in A$ let $M(x, \cdot)$ be the vector indexed by V whose entry at b is $\mathbb{P}(o \leftrightarrow A, x \leftrightarrow b)$, and let $\mathbf{w}$ be the vector indexed by V whose entry at b is $\mathbb{P}(o \leftrightarrow b, o \leftrightarrow A)$. Let $\mathcal{N}$ be the set of all sums of a convex combination of the vectors $M(x, \cdot)$ and a vector with nonnegative entries. The set $\mathcal{N}$ is convex, and it is closed because it is the sum of a compact set and a closed set in finite dimension.

Suppose $\mathbf{w} \notin \mathcal{N}$. Strict separation of $\mathbf{w}$ from $\mathcal{N}$ gives a vector y indexed by V and a real number t with $\langle y, \mathbf{w} \rangle < t \leq \langle y, \mathbf{z} \rangle$ for every $\mathbf{z} \in \mathcal{N}$. Fix $b \in V$ and fix a point $\mathbf{z}$ of the nonempty set $\mathcal{N}$. Increasing the coordinate b of $\mathbf{z}$ by a positive integer n and leaving the others alone produces another point of $\mathcal{N}$, so $\langle y, \mathbf{z} \rangle + n y_b \geq t$ for every n , which forces $y_b \geq 0$.

Lemma 6.1 therefore gives a vertex $a \in A$ with $\langle y, M(a, \cdot) \rangle \leq \langle y, \mathbf{w} \rangle$. But $M(a, \cdot)$ lies in $\mathcal{N}$, so separation gives $\langle y, M(a, \cdot) \rangle \geq t > \langle y, \mathbf{w} \rangle$, a contradiction. Hence $\mathbf{w}$ lies in $\mathcal{N}$: there are $c_x \geq 0$ with $\sum_{x \in A} c_x = 1$ and a vector with nonnegative entries whose sum with $\sum_{x \in A} c_x M(x, \cdot)$ is $\mathbf{w}$. Reading this coordinate by coordinate gives the assertion. $\square$

7 A Set of Vertices in Place of One, and Conjecture 6

This section replaces the single vertex o of Theorem 5.2 by a finite set of vertices, and deduces Conjecture 2.5. For a finite set R of vertices write $C_R = \bigcup_{s \in R} C_s$, so that $\{R \leftrightarrow B\}$ is the event $\{C_R \cap B \neq \emptyset\}$ and $\{R \nleftrightarrow B\}$ its complement. Nothing is assumed about the vertices of R , and C_R need not be connected.

Theorem 7.1. *Let A be a nonempty finite set of vertices, let $a \in A$, let R be a nonempty finite set of vertices, and let F be an increasing function of subsets of V with $\mathbb{E}[F(C_a)] \leq \mathbb{E}[F(C_x)]$ for every $x \in A$. Then*

$$\mathbb{E}[F(C_R) - F(C_a); R \nleftrightarrow a, R \leftrightarrow A \setminus \{a\}] \geq 0. \quad (22)$$

Section 7.1 deduces Conjecture 2.5 from it. Section 7.2 gives the proof of Theorem 7.1 as a chain of nine steps. Each step states its inequality exactly and describes the argument; the full arguments are the theorems named in Section 10.

7.1 Forcing the Pair Open

Two facts make the passage to the measure with e forced open possible: the union of the clusters of the two endpoints of e does not see the state of e , and the mean of a function blind to that state is the same under both measures.

Lemma 7.2. *Let v and w be distinct vertices, let e be the pair $\{v, w\}$, let a be a vertex, and let ω be a configuration in which e is closed. Then $C_{\{v, w\}}$ takes the same value in ω and in the configuration obtained from ω by opening e ; the same holds for the events $\{\{v, w\} \leftrightarrow B\}$ and $\{\{v, w\} \leftrightarrow B\}$ for every set B of vertices, and, on $\{\{v, w\} \leftrightarrow a\}$, for the cluster C_a .*

Proof. Opening e can only enlarge clusters, so $C_{\{v, w\}}$ can only grow. Conversely, a path in the enlarged configuration from v or w to a vertex u that uses e splits at its uses of e into segments of ω , each starting at v or at w ; the last segment exhibits u in $C_v \cup C_w$ computed in ω . Hence $C_{\{v, w\}}$ is unchanged, and so are the events determined by it. If $\{v, w\} \leftrightarrow a$ in ω , then no path of ω from a meets $\{v, w\}$, so no path from a in the enlarged configuration uses e , and C_a is unchanged as well. $\square$

Lemma 7.3. *Let e be a pair and let φ be a real function of configurations that takes the same value on a configuration in which e is closed and on the configuration obtained from it by opening e . Then the mean of φ under $\mathbb{P}^e$ equals its mean under $\mathbb{P}$.*

Proof. Condition on the states of all the pairs other than e . Under $\mathbb{P}$ and under $\mathbb{P}^e$ those states have the same joint law, since only the weight of e differs. Given them, the hypothesis says that φ takes one and the same value whether e is open or closed. The two conditional means agree, and averaging gives the assertion. $\square$

Theorem 7.4. *Let A be a nonempty finite set of vertices, let b be a vertex, let $a \in A$ minimise $\mathbb{P}(x \leftrightarrow b)$ over $x \in A$, and let $e = \{v, w\}$ be a pair. Then*

$$\mathbb{P}^e(a \leftrightarrow b, v \leftrightarrow A) \leq \mathbb{P}^e(v \leftrightarrow b, v \leftrightarrow A). \quad (23)$$

Proof. Apply Theorem 7.1 with $R = \{v, w\}$ and $F = \mathbf{1}\{b \in \cdot\}$, for which $\mathbb{E}[F(C_x)] = \mathbb{P}(x \leftrightarrow b)$, so that the hypothesis on a is exactly the minimality assumed here. By Lemmas 7.2 and 7.3 the left side of (22) has the same value under $\mathbb{P}$ and under $\mathbb{P}^e$, since the integrand and the two events defining the range of integration are unchanged by opening e on that range. Hence

$$\mathbb{E}^e[F(C_R) - F(C_a); R \leftrightarrow a, R \leftrightarrow A \setminus \{a\}] \geq 0, \quad R = \{v, w\}.$$

Under $\mathbb{P}^e$ the pair e is open with probability one, so almost surely $C_v = C_w = C_R$, the event $\{R \leftrightarrow a\}$ is $\{v \leftrightarrow a\}$ and the event $\{R \leftrightarrow A \setminus \{a\}\}$ is $\{v \leftrightarrow A \setminus \{a\}\}$. Split $\{v \leftrightarrow A\}$ into $\{v \leftrightarrow a\}$, on which $C_v = C_a$ and the integrand vanishes, and $\{v \leftrightarrow a\} \cap \{v \leftrightarrow A \setminus \{a\}\}$. The last display becomes $\mathbb{E}^e[\mathbf{1}\{b \in C_v\} - \mathbf{1}\{b \in C_a\}; v \leftrightarrow A] \geq 0$, which is (23). $\square$

Conjecture 2.5 follows. The events $\{a \leftrightarrow b\}$ and $\{v \leftrightarrow A\}$ are increasing, so the Harris inequality and (23) give

$$\mathbb{P}^e(v \leftrightarrow A)\mathbb{P}^e(a \leftrightarrow b) \leq \mathbb{P}^e(a \leftrightarrow b, v \leftrightarrow A) \leq \mathbb{P}^e(v \leftrightarrow b, v \leftrightarrow A) \leq \mathbb{P}^e(v \leftrightarrow b),$$

which is (3). Its hypothesis (2) holds automatically: it is the case of Conjecture 2.1 with x in place of o .

7.2 The Nine Steps

Throughout this subsection fix a vertex s , a set Y of vertices with $s \notin Y$, and an increasing function g of sets of pairs, and write $D = \{s \leftrightarrow Y\}$. For finite sets R , X and B of vertices put

$$H(R; X, B) = \{R \leftrightarrow B\} \cap \{R \leftrightarrow X\}, \quad (24)$$

the event that C_R misses B and meets X . The event $\{R \leftrightarrow s\}$ does not imply $\{R \leftrightarrow Y\}$ even when $\{s \leftrightarrow Y\}$ holds, because a component of C_R other than the one containing s may meet Y . The two conditions must therefore be carried together. For a single vertex, however, $H(\{u\}; \{s\}, Y) = D \cap \{s \leftrightarrow u\}$, since $u \leftrightarrow s$ makes the clusters of u and s equal. Set

$$\Gamma(R) = \mathbb{P}(D) \mathbb{E}[g(C_s); H(R; \{s\}, Y)] - \mathbb{E}[g(C_s); D] \mathbb{P}(H(R; \{s\}, Y)), \quad (25)$$

which at $R = \{u\}$ is $\mathbb{P}(D)^2$ times the conditioned covariance of Theorem 3.3. The event $H(R; \{s\}, Y)$ is contained in D : on it s lies in C_R and C_R misses Y , so $C_s \subseteq C_R$ misses Y .

The functional (8) is now applied to functions of a set argument. Fix distinct vertices $z_1, \dots, z_\ell$ and a further vertex v , all outside $\{s\} \cup Y$ and distinct from each other, and recall the sets B_j of (7). Put

$$\pi_j(R) = \frac{\mathbb{P}(H(R; \{z_j\}, B_j))}{\mathbb{P}(z_j \leftrightarrow B_j)}, \quad \tau(R) = \frac{\mathbb{P}(H(R; \{v\}, B_{\ell+1}))}{\mathbb{P}(v \leftrightarrow B_{\ell+1})}. \quad (26)$$

For a real function q of finite vertex sets put $q^0 = q$, then $q^j(R) = q^{j-1}(R) - \pi_j(R)q^{j-1}(\{z_j\})$ for $1 \leq j \leq \ell$, and finally, for a prescribed set R_0 ,

$$\mathfrak{M}[q] = q^\ell(R_0) - \tau(R_0)q^\ell(\{v\}). \quad (27)$$

At a singleton R_0 the numbers (26) are the constants (7), and $\mathfrak{M}[\Gamma] \geq 0$ is Theorem 3.3. The goal of the first seven steps is the same conclusion for a set R_0 with at least two elements.

Step 1. What the discounting does to a function of sets. There are real numbers, depending only on $z_1, \dots, z_\ell$, on v and on R_0 , such that $\mathfrak{M}[q]$ is $q(R_0)$ plus the corresponding linear combination of the values $q(\{u\})$ for $u \in V$, for every q . This follows by induction on ℓ : applying the first discounting step turns $\mathfrak{M}$ into the functional for $z_2, \dots, z_\ell$ applied to $R \mapsto q(R) - \pi_1(R)q(\{z_1\})$, and every term the subtraction produces is a multiple of $q(\{z_1\})$. Only R_0 enters with a coefficient not attached to a singleton, and that coefficient is one. This is what makes the one-sided estimates below usable.

Step 2. A set reaching a vertex, as a mean over the cluster of that vertex. For a set U of vertices write $\mathbb{P}_U$ and $\mathbb{E}_U$ as in Section 3. Let X be a set of vertices contained in U and containing s , let $v \in U \setminus X$, and let $R \subseteq U$ be disjoint from $X \cup \{v\}$. For a set K of vertices put

$$\alpha_{R,B}(K) = \mathbf{1}\{K \cap R \neq \emptyset\} \mathbb{P}_{U \setminus K}(R \setminus K \leftrightarrow B). \quad (28)$$

Then $\alpha_{R,B}(K)$ is increasing in K and decreasing in B , and

$$\mathbb{E}_U[\alpha_{R,B}(C_v); v \leftrightarrow B] = \mathbb{P}_U(R \leftrightarrow B, R \leftrightarrow v). \quad (29)$$

The identity follows by summing over the values of C_v : on $\{v \leftrightarrow B\}$ the set $K = C_v$ misses B , the event $\{R \leftrightarrow v\}$ is $\{K \cap R \neq \emptyset\}$, and the pairs with both endpoints outside K retain their original weights. Applying Theorem 3.1 inside U to the vertex v , to the two sets $X \cup N$ and $X \cup N_1$, and to the functions $\alpha_{R, X \cup N}$ and the constant one, then gives, for disjoint N and N_1 ,

$$\mathbb{P}_U(R \leftrightarrow X \cup N, R \leftrightarrow v) \mathbb{P}_U(v \leftrightarrow X \cup N_1) \leq \mathbb{P}_U(v \leftrightarrow X \cup N \cup N_1) \mathbb{P}_U(R \leftrightarrow X, R \leftrightarrow v). \quad (30)$$

The two avoided sets meet in X and their union is $X \cup N \cup N_1$.

Step 3. Removing the disjointness, by the Ahlswede and Daykin inequality. Write $\varrho(W)$ for $\mathbb{P}_W(R \leftrightarrow X, R \leftrightarrow v) / \mathbb{P}_W(v \leftrightarrow X)$ when the denominator is positive, and $\varrho(W) = 0$ otherwise. Let η be a nonnegative function of subsets of U that vanishes on sets not containing v , and put

$$\phi(N) = \mathbb{E}_U[\mathbf{1}\{s \leftrightarrow N\} \eta(U \setminus C_N)], \quad \chi(N) = \mathbb{E}_U[\mathbf{1}\{s \leftrightarrow N\} \mathbf{1}\{R \leftrightarrow N\} \varrho(U \setminus C_N) \eta(U \setminus C_N)].$$

Assume that for every $W \subseteq U$ and every $Q \subseteq W$ the function ϕ formed inside W satisfies $\phi(Q)\mathbb{P}_W(v \leftrightarrow X) \leq \mathbb{P}_W(v \leftrightarrow X \cup Q)\eta(W)$, and that $Q \mapsto \phi(Q)$ is decreasing in every such W . Then, for all subsets N and N_1 of U ,

$$\mathbb{P}_U(R \leftrightarrow X \cup N, R \leftrightarrow v) \phi(N_1) \leq \mathbb{P}_U(v \leftrightarrow X \cup N \cup N_1) \chi(N \cap N_1). \quad (31)$$

The proof is an induction on the number of elements of U . When N and N_1 are disjoint, the assertion is (30) multiplied by the assumed inequality at N_1 , together with $\chi(\emptyset) = \varrho(U)\eta(U)$. Otherwise the intersection $Z = N \cap N_1$ is removed and the set of vertices joined to Z by an open pair is exposed. That set has a product law, and each of the four quantities is its average over that set of the corresponding quantity inside $U \setminus Z$. For two values of the exposed set the induction hypothesis inside $U \setminus Z$ gives the pointwise inequality between the four integrands, with the union and the intersection of the two values in the right places. The Ahlswede and Daykin inequality then averages it.

Step 4. A set of vertices against a single one. Let $\eta_R(U) = \text{Cov}_U(g(\mathcal{C}_s), \mathbf{1}\{R \leftrightarrow X\})$ and let $\eta_v(U)$ be as in (11). Then $\eta_v(U) \geq 0$ and

$$\varrho(U) \eta_v(U) \leq \eta_R(U). \quad (32)$$

Write J for $\{R \leftrightarrow X\} \cap \{R \leftrightarrow v\}$. The union of $\{R \leftrightarrow X\}$ and J is increasing, so Harris gives $\eta_R \geq \mathbb{E}_U[g(\mathcal{C}_s)]\mathbb{P}_U(J) - \mathbb{E}_U[g(\mathcal{C}_s); J]$. The function $g(\mathcal{C}_s)$ is increasing in $\mathcal{C}_X$ and does not depend on $\mathcal{C}_v$, while $\mathbf{1}_J$ is decreasing in $\mathcal{C}_X$ and increasing in $\mathcal{C}_v$. Hence (5) with $S_1 = X$ and $S_2 = \{v\}$ gives $\mathbb{P}_U(v \leftrightarrow X)\mathbb{E}_U[g(\mathcal{C}_s); J] \leq \mathbb{P}_U(J)\mathbb{E}_U[g(\mathcal{C}_s); v \leftrightarrow X]$, and substituting the definitions gives (32).

Step 5. What survives once the cluster of Y is exposed. Let $X = \{s\} \cup \{z_1, \dots, z_\ell\}$, let v lie outside $X \cup Y$, and let R_0 be disjoint from $X \cup Y \cup \{v\}$. Write $U = V \setminus C_Y$ and

$$\mathcal{H} := \mathbb{E}[\mathbf{1}_D \mathbf{1}\{R_0 \leftrightarrow Y\} \eta_{R_0}(U)] - \tau(R_0) \mathbb{E}[\mathbf{1}_D \eta_v(U)].$$

If every pair weight lies strictly between zero and one, then $\mathcal{H} \geq 0$. Apply (31) with $U = V$, with $R = R_0$, with $N = N_1 = Y$ and with $\eta = \eta_v$, whose two hypotheses are (12) and the monotonicity recorded with it. Dividing by $\mathbb{P}(v \leftrightarrow X \cup Y)$, which is positive, and recognising $\tau(R_0)$ as the resulting quotient, gives $\tau(R_0)\mathbb{E}[\mathbf{1}_D \eta_v(U)] \leq \mathbb{E}[\mathbf{1}_D \mathbf{1}\{R_0 \leftrightarrow Y\} \varrho(U) \eta_v(U)]$. Then (32) inside U bounds the integrand.

Step 6. Exploring the cluster of a set, not of a vertex. Let z be a vertex and let B contain $\{s\} \cup Y$ but not z , and abbreviate $H_R = H(R; \{z\}, B)$ and $E = \{z \leftrightarrow B\}$, with $\mathbb{P}(E) > 0$. Summing over the values of $\mathcal{C}_R$ gives $\mathbb{E}[\Theta; H_R] = -\mathbb{E}[\Psi(C_R); H_R]$: on H_R the set C_R misses B and every pair with exactly one endpoint in C_R is closed, so the conditional mean of Θ is $-\Psi(C_R)$. On H_R at least one vertex of R is joined to z , so $C_z \subseteq C_R$, and therefore

$$\Delta(R) = \mathbb{E}[\Psi(C_R) - \Psi(C_z); H_R] \geq 0,$$

with $\Delta(\{u\}) = 0$ for every vertex u , since $C_{\{u\}} = C_z$ on $H_{\{u\}}$. Let $\Gamma_{z,B}$ denote (25) formed with z , with B and with the increasing function $\mathcal{C} \mapsto \Psi(\{z\} \cup \text{span } \mathcal{C})$. Expanding $\Gamma_{z,B}$, then using the displayed identity, the definition of Δ and (10), gives

$$\mathbb{E}[\Theta; H_R] - \pi(R) \mathbb{E}[\Theta; E] = -\frac{\Gamma_{z,B}(R)}{\mathbb{P}(E)} - \Delta(R), \quad \pi(R) = \frac{\mathbb{P}(H_R)}{\mathbb{P}(E)}, \quad (33)$$

and hence, dropping $\Delta(R) \geq 0$,

$$\mathbb{E}[\Theta; H_R] - \pi(R) \mathbb{E}[\Theta; E] \leq -\frac{\Gamma_{z,B}(R)}{\mathbb{P}(E)}, \quad (34)$$

with equality whenever R is a single vertex.

Step 7. Peeling the auxiliary vertices, and the conclusion for a set. Put $\mathcal{W} = \mathbb{E}[\Theta \cdot \mathfrak{M}[R \mapsto \mathbf{1}_{H(R; \{s\}, Y)}]]$, where $\mathfrak{M}$ acts on the argument R with $\mathcal{C}_s$ and $\mathcal{C}_Y$ held fixed. For $1 \leq j \leq \ell$ let $\mathfrak{M}_j$ be the functional (27) built from $z_{j+1}, \dots, z_\ell$, from v and from R_0 with base set B_{j+1} . Then

$$\mathcal{W} = \mathcal{H} + \sum_{j=1}^{\ell} \frac{\mathfrak{M}_j[\Gamma_{z_j, B_j}]}{\mathbb{P}(z_j \leftrightarrow B_j)} + \sum_{j=1}^{\ell} \Delta_j(R_0). \quad (35)$$

The identity comes from removing $z_1, \dots, z_\ell$ from the indicator one at a time. On a fixed value of $\mathcal{C}_Y$, the indicator that R misses $\mathcal{C}_Y$ and meets a set W splits, as W grows by one vertex z , into the indicator for the larger set, the two terms whose difference is the left side of (33), and a term that depends on the configuration only through $\mathcal{C}_Y$. The last contributes nothing, because $\mathbb{E}[\Theta \mid \mathcal{C}_Y] = 0$. By the first step the terms produced at the j th removal are evaluated at R_0 with coefficient one and at singletons otherwise, and Δ_j vanishes at singletons. After ℓ removals the indicator has grown to the one for $\{s\} \cup \{z_1, \dots, z_\ell\}$, and what is left of it is $\mathcal{H}$.

Consequently $\mathfrak{M}[\Gamma] \geq 0$ whenever every pair weight lies strictly between zero and one and R_0 has at least two elements. The proof is an induction on ℓ which proves $\mathcal{W} \geq 0$ and $\mathfrak{M}[\Gamma] \geq 0$ together. Nonnegativity of $\mathcal{W}$ is (35) together with Step 5, the induction hypothesis at a shorter list, and $\Delta_j \geq 0$. Nonnegativity of $\mathfrak{M}[\Gamma]$ then follows by writing $\mathfrak{M}[\Gamma]$ as a conditioned covariance on D against a test function of $(\mathcal{C}_s, \mathcal{C}_Y)$ that is decreasing in the second, and splitting that covariance by alternately conditioning on $\mathcal{C}_Y$ and on $\mathcal{C}_s$. The first half is $\mathcal{W}$; the second is nonnegative by Harris for two decreasing functions; the remainder is the same covariance with the first argument replaced by a two-step average, and those averages converge to a constant on D .

Step 8. Removing the vertices of T one at a time. Keep Y, F, T, r and the numbers m_x of Section 4. For a finite set R of vertices put

$$J_x(R) = \{R \leftrightarrow Y\} \cap \{R \leftrightarrow x\} \cap \bigcap_{y \in T, r(y) < r(x)} \{R \leftrightarrow y\},$$

$$\Lambda_R(T) = \mathbb{E}[F(C_R); R \leftrightarrow Y, R \leftrightarrow T] - \sum_{x \in T} \mathbb{P}(J_x(R)) m_x.$$

For $R = \{u\}$ these are (14) and (15).

Theorem 7.5. *With V, Y, F, T and r as in Section 4, let R be a nonempty finite set of vertices disjoint from $Y \cup T$. Then $\Lambda_R(T) \geq 0$.*

The proof follows that of Theorem 4.1, with $\mathfrak{M}[\Gamma] \geq 0$ of Step 7 in place of Theorem 3.3. Let $k \in T$ have the largest value of r , put $T_0 = T \setminus \{k\}$ and $H_R = H(R; \{k\}, Y \cup T)$. Adding k creates the single new event H_R , so

$$\Lambda_R(T \cup \{k\}) - \Lambda_R(T) = \mathbb{E}[F(C_R) - m_k; H_R], \quad (36)$$

which is at least $\mathbb{E}[F(C_k) - m_k; H_R]$ because $C_k \subseteq C_R$ on H_R , with equality when R is a single vertex. With $\gamma_k = m_k \mathbb{P}(k \leftrightarrow Y \cup T) - \mathbb{E}[F(C_k); k \leftrightarrow Y \cup T]$, expanding both sides gives

$$\mathbb{P}(k \leftrightarrow Y \cup T) \mathbb{E}[F(C_k) - m_k; H_R] = \Gamma_{k, Y \cup T}(R) - \gamma_k \mathbb{P}(H_R), \quad (37)$$

and $\gamma_k \leq \Lambda_{\{k\}}(T)$ exactly as in (20). Two inductions on the cardinality of T then run as before. The first proves $\mathfrak{M}[R_1 \mapsto \Lambda_{R_1}(T)] \geq 0$, using Step 7 twice: once for the increasing function $\mathcal{C} \mapsto F(\{k\} \cup \text{span } \mathcal{C})$, and once for $\mathcal{C} \mapsto \mathbf{1}\{\mathcal{C} \neq \emptyset\}$, which shows that the value of $\mathfrak{M}$ at π is nonnegative. Its case $\ell = 0$ with $v = k$ is

$$\mathbb{P}(H(R; \{k\}, Y \cup T_0)) \Lambda_{\{k\}}(T_0) \leq \mathbb{P}(k \leftrightarrow Y \cup T_0) \Lambda_R(T_0). \quad (38)$$

The second induction combines (38) with (6) applied to $\mathcal{C} \mapsto F(\{k\} \cup \text{span } \mathcal{C})$ and $\mathcal{C} \mapsto \alpha_{R, Y \cup T_0}(\{k\} \cup \text{span } \mathcal{C})$, whose mean is computed by (29). The passage to weights in $[0, 1]$ uses the analogue of (16) for $\Lambda_R(T)$, which does not refer to r and is continuous wherever every $\mathbb{P}(x \leftrightarrow Y)$ is positive.

Step 9. Passing to Φ_a , and the proof of Theorem 7.1. Two identities relate F to Φ_a when a set of vertices replaces a single one. Let R be a nonempty finite set of vertices, let a be a vertex and let $\mathcal{S}$ be a family of sets of pairs. Then

$$\mathbb{E}[F(C_R) - F(C_a); R \leftrightarrow a, C_R \in \mathcal{S}] = \mathbb{E}[\Phi_a(C_R); R \leftrightarrow a, C_R \in \mathcal{S}], \quad (39)$$

by the argument of Lemma 5.1(i) with $\mathcal{C}_R$ in place of $\mathcal{C}_o$. If moreover some vertex $s \in R$ has $\mathbb{E}[F(C_a)] \leq \mathbb{E}[F(C_s)]$, then

$$\mathbb{E}[F(C_R) - F(C_a); R \leftrightarrow a] \geq 0. \quad (40)$$

For (40), take $\alpha_{R,\{a\}}$ of (28) with $U = V$: at the cluster of s its value is the probability that no vertex of $R \setminus \{s\}$ reaches a once every pair meeting that cluster is closed. It is increasing, and (29) turns an integral over $\{R \leftrightarrow a\}$ of a function of $\mathcal{C}_s$ into the integral over $\{s \leftrightarrow a\}$ of that function times $\alpha_{R,\{a\}}$. Now apply (6) with the vertex s , with $X = \{a\}$ and with the two increasing functions $\mathcal{C} \mapsto \Phi_a(\{s\} \cup \text{span } \mathcal{C})$ and $\mathcal{C} \mapsto \alpha_{R,\{a\}}(\{s\} \cup \text{span } \mathcal{C})$. The first factor $\mathbb{E}[\Phi_a(C_s); s \leftrightarrow a]$ is $\mathbb{E}[F(C_s)] - \mathbb{E}[F(C_a)] \geq 0$ by Lemma 5.1(ii), so $\mathbb{E}[\Phi_a(C_s); R \leftrightarrow a] \geq 0$; then $C_s \subseteq C_R$ and (39) give (40).

Theorem 7.1 now follows. Put $T = A \setminus \{a\}$. If $a \in R$, then $\{R \leftrightarrow a\}$ is empty and both sides of (22) vanish. If R meets T , choose $s \in R \cap T$; then $\{R \leftrightarrow T\}$ is the sure event and (40) applies. Otherwise R is disjoint from A . Let T_1 be the set of $x \in T$ with $\mathbb{P}(x \leftrightarrow a) > 0$, and apply Theorem 7.5 with $Y = \{a\}$, with T_1 , with Φ_a and with R . By Lemma 5.1(ii) the number (13) attached to $x \in T_1$ is $(\mathbb{E}[F(C_x)] - \mathbb{E}[F(C_a)])/\mathbb{P}(x \leftrightarrow a) \geq 0$, so $\mathbb{E}[\Phi_a(C_R); R \leftrightarrow a, R \leftrightarrow T_1] \geq 0$; and (39) converts this into (22) with T_1 in place of $A \setminus \{a\}$. Finally a configuration in $\{R \leftrightarrow a\} \cap \{R \leftrightarrow T\}$ but not in $\{R \leftrightarrow T_1\}$ has R joined to at least one vertex x of $T \setminus T_1$ while missing a , hence lies in the null event $\{x \leftrightarrow a\}$. A finite union of null events is null, so the two set integrals agree.

8 Deleting the Edges at the Starting Vertex

Question 2.9 selects a by a probability computed with every pair incident to o closed, and then asks for (4) in the original measure. Exposing the pairs at o turns the cluster of o into the union of the clusters of a random set of vertices, and Theorem 7.1 applies to each value of that set.

Theorem 8.1. *Let A be a nonempty finite set of vertices, let o and b be vertices, and let $a \in A$ minimise $\mathbb{P}_*(x \leftrightarrow b)$ over $x \in A$. Then (4) holds. In particular Question 2.9 has an affirmative answer.*

Proof. Condition on the states of the pairs incident to o . For such a state ζ let R_ζ be the union of $\{o\}$ and the set of vertices $u \neq o$ with $\{o, u\}$ open in ζ . The pairs not incident to o have exactly their law under $\mathbb{P}_*$, and they are independent of ζ .

Under $\mathbb{P}$ a path leaving o starts with a pair of ζ and continues among the pairs not incident to o , possibly returning to o and leaving again. Hence, pointwise in the remaining configuration,

$$C_o = C_{R_\zeta}, \quad \{o \leftrightarrow A\} = \{R_\zeta \leftrightarrow A\}, \quad (41)$$

the right sides being computed under $\mathbb{P}_*$, for which o is isolated and therefore contributes the single vertex o to the union. Moreover, on $\{o \leftrightarrow a\}$ no path from a can reach o , so no such path uses a pair of ζ , and C_a is the same under both measures.

Put $F = \mathbf{1}\{b \in \cdot\}$. On $\{o \leftrightarrow a\}$ the clusters of o and a coincide, so

$$\mathbb{P}(o \leftrightarrow b, o \leftrightarrow A) - \mathbb{P}(a \leftrightarrow b, o \leftrightarrow A) = \mathbb{E}[F(C_o) - F(C_a); o \leftrightarrow a, o \leftrightarrow A \setminus \{a\}],$$

and by (41) the right side equals

$$\sum_{\zeta} \mathbb{P}(\zeta) \mathbb{E}_*[F(C_{R_\zeta}) - F(C_a); R_\zeta \leftrightarrow a, R_\zeta \leftrightarrow A \setminus \{a\}],$$

the events and clusters inside the expectation being computed under $\mathbb{P}_*$. The hypothesis on a says $\mathbb{E}_*[F(C_a)] \leq \mathbb{E}_*[F(C_x)]$ for every $x \in A$, so each summand is nonnegative by Theorem 7.1 applied under $\mathbb{P}_*$ with the set R_ζ . Averaging gives (4). No division is used, so states ζ of probability zero and coincidences among o , a , b and the vertices of A are covered. $\square$

9 A Four-Vertex Example, and Sets of Two Vertices

Question 2.8 selects a by the probability of the joint event that a reaches b and that o misses A . Write ρ_x for that probability, so that

$$\rho_x = \mathbb{P}(x \leftrightarrow b, o \nleftrightarrow A) \quad \text{for } x \in A.$$

When several vertices of A realise the minimum of ρ , the reading in which every one of them satisfies (4) is false.

Proposition 9.1. *There are a graph with four vertices, weights in $[0, 1]$, a set A with two elements and vertices o and b outside A such that both elements of A minimise ρ while (4) fails for one of them.*

Proof. Let $V = \{o, a_1, a_2, b\}$ and $A = \{a_1, a_2\}$. Give the pairs $\{o, a_1\}$ and $\{a_2, b\}$ weight one and the pair $\{a_1, a_2\}$ weight one half, and give every other pair weight zero. Since $\{o, a_1\}$ is open with probability one and $a_1 \in A$, the event $\{o \leftrightarrow A\}$ has probability one, so $\rho_{a_1} = \rho_{a_2} = 0$ and both vertices minimise. The event $\{a_2 \leftrightarrow b\}$ has probability one, so $\mathbb{P}(o \leftrightarrow A, a_2 \leftrightarrow b) = 1$. The event $\{o \leftrightarrow b\}$ requires the pair $\{a_1, a_2\}$ to be open, so $\mathbb{P}(o \leftrightarrow b, o \leftrightarrow A) = \frac{1}{2}$. Thus (4) fails for $a = a_2$. $\square$

The minimised quantity vanishes at both vertices of A there. Nothing like it can happen when A has at most two elements. Two preparations are needed. The first removes from (4) the part of $\{o \leftrightarrow A\}$ on which its two sides agree.

Lemma 9.2. *Let A be a nonempty finite set of vertices, let $a \in A$, and let b and o be vertices. Then*

$$\mathbb{P}(o \leftrightarrow A, a \leftrightarrow b) - \mathbb{P}(o \leftrightarrow b, o \leftrightarrow A) = \mathbb{P}(a \leftrightarrow b, o \nleftrightarrow a, o \leftrightarrow A \setminus \{a\}) - \mathbb{P}(o \leftrightarrow b, o \nleftrightarrow a, o \leftrightarrow A \setminus \{a\}).$$

Proof. Split $\{o \leftrightarrow A\}$ into $\{o \leftrightarrow a\}$ and $\{o \nleftrightarrow a\} \cap \{o \leftrightarrow A \setminus \{a\}\}$. On $\{o \leftrightarrow a\}$ the clusters of o and a coincide, so the events $\{a \leftrightarrow b\}$ and $\{o \leftrightarrow b\}$ agree there and the two contributions from that part cancel. $\square$

The second preparation is a four-product form of Theorem 3.2.

Lemma 9.3. *Let S_1 and S_2 be sets of vertices, let g_3 and g_4 be nonnegative real functions of two sets of pairs, each increasing in the first argument and decreasing in the second, and let h_1 and h_2 be nonnegative real functions of two sets of pairs, each decreasing in the first argument and increasing in the second. Then, with every function evaluated at (C_{S_1}, C_{S_2}) and every expectation restricted to $\{S_1 \leftrightarrow S_2\}$,*

$$\mathbb{E}[g_3 h_1] \mathbb{E}[g_4 h_2] \leq \mathbb{E}[g_3 g_4] \mathbb{E}[h_1 h_2].$$

Proof. Write D for $\{S_1 \leftrightarrow S_2\}$. If $\mathbb{P}(D) = 0$, then both sides vanish, so assume $\mathbb{P}(D) > 0$. Theorem 3.2 applies to the pair g_3, g_4 and, after exchanging S_1 and S_2 , to the pair h_1, h_2 , giving

$$\mathbb{E}[g_3] \mathbb{E}[g_4] \leq \mathbb{P}(D) \mathbb{E}[g_3 g_4], \quad \mathbb{E}[h_1] \mathbb{E}[h_2] \leq \mathbb{P}(D) \mathbb{E}[h_1 h_2].$$

Meanwhile (5) applies to the pair g_3, h_1 and to the pair g_4, h_2 , giving

$$\mathbb{P}(D) \mathbb{E}[g_3 h_1] \leq \mathbb{E}[g_3] \mathbb{E}[h_1], \quad \mathbb{P}(D) \mathbb{E}[g_4 h_2] \leq \mathbb{E}[g_4] \mathbb{E}[h_2].$$

All eight quantities are nonnegative. Multiplying the two inequalities of the second display and bounding the result by the product of the two of the first gives $\mathbb{P}(D)^2 \mathbb{E}[g_3 h_1] \mathbb{E}[g_4 h_2] \leq \mathbb{P}(D)^2 \mathbb{E}[g_3 g_4] \mathbb{E}[h_1 h_2]$. Divide by $\mathbb{P}(D)^2$. $\square$

Theorem 9.4. *Let o and b be vertices and let A be a nonempty set of vertices with at most two elements. If $a \in A$ is the unique vertex of A realising the minimum of ρ , then (4) holds. If moreover $\mathbb{P}(o \nleftrightarrow A) > 0$, then (4) holds for every $a \in A$ realising that minimum.*

Proof. We argue in four steps: a set with one element; a set with two elements at which the minimum is strict; the first assertion; and the tied minima of the second assertion.

If $A = \{a\}$, then on $\{o \leftrightarrow a\}$ the clusters of o and a coincide, so $\mathbb{P}(o \leftrightarrow A, a \leftrightarrow b) = \mathbb{P}(o \leftrightarrow a, o \leftrightarrow b) = \mathbb{P}(o \leftrightarrow b, o \leftrightarrow A)$, and both assertions hold with equality.

Next let $A = \{a, x\}$ with $\rho_a < \rho_x$. By Lemma 9.2, and because the clusters of o and x coincide on $\{o \leftrightarrow x\}$, the inequality (4) to be proved reads

$$\mathbb{P}(a \leftrightarrow b, o \nleftrightarrow a, o \leftrightarrow x) \leq \mathbb{P}(x \leftrightarrow b, o \nleftrightarrow a, o \leftrightarrow x). \quad (42)$$

Apply Lemma 9.3 with $S_1 = \{a\}$ and $S_2 = \{o, x\}$, and write D for the event $\{a \nleftrightarrow o, a \nleftrightarrow x\}$. Take for g_3 and g_4 the indicators of $\{a \leftrightarrow b\}$ and of $\{o \nleftrightarrow x\}$, and for h_1 and h_2 the indicators of $\{o \leftrightarrow x\}$ and of $\{x \leftrightarrow b\}$. The event $\{a \leftrightarrow b\}$ is determined by the open pairs lying on paths from a and increases with them. The events $\{o \leftrightarrow x\}$ and $\{x \leftrightarrow b\}$ are determined by the open pairs lying on paths from $\{o, x\}$ and increase with them. Hence g_3 and g_4 are increasing in the first argument and decreasing in the second, and h_1 and h_2 are of the opposite sort.

We identify the four expectations. On $\{o \leftrightarrow x\}$ the clusters of o and x coincide, so $a \nleftrightarrow o$ and $a \nleftrightarrow x$ are the same condition there and $D \cap \{o \leftrightarrow x\} = \{o \nleftrightarrow a, o \leftrightarrow x\}$; therefore

$$\mathbb{E}[g_3 h_1; D] = \mathbb{P}(a \leftrightarrow b, o \nleftrightarrow a, o \leftrightarrow x), \quad \mathbb{E}[h_1 h_2; D] = \mathbb{P}(x \leftrightarrow b, o \nleftrightarrow a, o \leftrightarrow x).$$

On the other hand $D \cap \{o \nleftrightarrow x\} = \{o \nleftrightarrow A\} \cap \{a \nleftrightarrow x\}$, and two vertices joined to b are joined to each other, so on that event $\{a \leftrightarrow b\}$ forces $x \leftrightarrow b$ and $\{x \leftrightarrow b\}$ forces $a \leftrightarrow b$, and conversely. Therefore

$$\mathbb{E}[g_3 g_4; D] = \mathbb{P}(o \nleftrightarrow A, a \leftrightarrow b, x \leftrightarrow b), \quad \mathbb{E}[g_4 h_2; D] = \mathbb{P}(o \nleftrightarrow A, x \leftrightarrow b, a \leftrightarrow b),$$

and Lemma 9.3 reads

$$\begin{aligned} \mathbb{P}(a \leftrightarrow b, o \nleftrightarrow a, o \leftrightarrow x) \mathbb{P}(o \nleftrightarrow A, x \leftrightarrow b, a \leftrightarrow b) \\ \leq \mathbb{P}(o \nleftrightarrow A, a \leftrightarrow b, x \leftrightarrow b) \mathbb{P}(x \leftrightarrow b, o \nleftrightarrow a, o \leftrightarrow x). \end{aligned} \quad (43)$$

Splitting $\{o \nleftrightarrow A, a \leftrightarrow b\}$ and $\{o \nleftrightarrow A, x \leftrightarrow b\}$ according as the other vertex of A reaches b ,

$$\rho_a - \mathbb{P}(o \nleftrightarrow A, a \leftrightarrow b, x \leftrightarrow b) = \mathbb{P}(o \nleftrightarrow A, a \leftrightarrow b, x \leftrightarrow b) = \rho_x - \mathbb{P}(o \nleftrightarrow A, x \leftrightarrow b, a \leftrightarrow b),$$

so $\rho_a < \rho_x$ gives $\mathbb{P}(o \nleftrightarrow A, a \leftrightarrow b, x \leftrightarrow b) < \mathbb{P}(o \nleftrightarrow A, x \leftrightarrow b, a \leftrightarrow b)$, and in particular the second of these numbers is positive. Bounding the first factor on the right of (43) by the second number and cancelling that positive factor gives (42).

For the first assertion, a set with one element is covered by the first step. If $A = \{a, x\}$ and a is the unique vertex realising the minimum, then $\rho_a < \rho_x$ and the second step applies.

For the second assertion, assume $\mathbb{P}(o \nleftrightarrow A) > 0$ and let a realise the minimum. Again only $A = \{a, x\}$ has to be treated, and only $\rho_a = \rho_x$.

Suppose first that $\rho_a = 0$. A configuration has positive probability precisely when it contains every pair of weight one and no pair of weight zero. The event $\{o \nleftrightarrow A\}$ is decreasing and has positive probability, so it contains the configuration whose open pairs are exactly those of weight one; that is, the component of o in the graph of the pairs of weight one misses A . Suppose $\mathbb{P}(a \leftrightarrow b, o \nleftrightarrow b) > 0$, and choose a configuration of positive probability in which $a \leftrightarrow b$ and $o \nleftrightarrow b$, together with a simple open path from a to b in it. Let ω be the configuration whose open pairs are those of weight one together with the pairs of that path. Then ω has positive probability, it is contained in the chosen configuration, and $a \leftrightarrow b$ in ω . Moreover $o \nleftrightarrow A$ in ω : a path in ω from o to a vertex of A either uses only pairs of weight one, which contradicts the statement about the component of o , or meets the chosen path, in which case $o \leftrightarrow b$ in ω and hence in the chosen configuration, contrary to its choice. Thus $\rho_a > 0$, a contradiction. Hence $\{a \leftrightarrow b\}$ is contained in $\{o \leftrightarrow b\}$ up to a null event, and when both hold the vertex o is joined to a through b , so $\{o \leftrightarrow A\}$ holds as well; therefore $\mathbb{P}(o \leftrightarrow A, a \leftrightarrow b) \leq \mathbb{P}(a \leftrightarrow b) \leq \mathbb{P}(o \leftrightarrow b, o \leftrightarrow A)$.

Suppose finally that $\rho_a > 0$, and fix q with $0 < q < 1$. Enlarge the vertex set by one vertex y , give the pair $\{y, a\}$ weight q and every other pair at y weight zero, leave every old weight unchanged, and replace A by $\{y, x\}$; write $\mathbb{P}_q$ for the enlarged measure. Condition on the state of $\{y, a\}$. If that pair is

open, then y is joined exactly to the vertices of the cluster of a , the connections among the old vertices are unchanged, and the events $\{o \leftrightarrow \{y, x\}\}$ and $\{y \leftrightarrow b\}$ are $\{o \leftrightarrow A\}$ and $\{a \leftrightarrow b\}$. If it is closed, then y is isolated, so $\{y \leftrightarrow b\}$ is empty and $\{o \leftrightarrow \{y, x\}\}$ is $\{o \leftrightarrow x\}$. Hence

$$\mathbb{P}_q(y \leftrightarrow b, o \leftrightarrow \{y, x\}) = q\rho_a, \quad \mathbb{P}_q(x \leftrightarrow b, o \leftrightarrow \{y, x\}) = q\rho_x + (1 - q)\mathbb{P}(x \leftrightarrow b, o \leftrightarrow x),$$

$$\mathbb{P}_q(o \leftrightarrow \{y, x\}, y \leftrightarrow b) = q\mathbb{P}(o \leftrightarrow A, a \leftrightarrow b),$$

$$\mathbb{P}_q(o \leftrightarrow b, o \leftrightarrow \{y, x\}) = q\mathbb{P}(o \leftrightarrow b, o \leftrightarrow A) + (1 - q)\mathbb{P}(o \leftrightarrow b, o \leftrightarrow x).$$

Since $\{o \leftrightarrow A\}$ is contained in $\{o \leftrightarrow x\}$, the number $\mathbb{P}(x \leftrightarrow b, o \leftrightarrow x)$ is at least ρ_x . The second of the two numbers in the first display is therefore at least $\rho_x = \rho_a > q\rho_a$, which is the first: the vertex y is the unique vertex of $\{y, x\}$ realising the minimum, in a set with two elements. By the first assertion,

$$q\mathbb{P}(o \leftrightarrow A, a \leftrightarrow b) \leq q\mathbb{P}(o \leftrightarrow b, o \leftrightarrow A) + (1 - q)\mathbb{P}(o \leftrightarrow b, o \leftrightarrow x).$$

Suppose $\delta = \mathbb{P}(o \leftrightarrow A, a \leftrightarrow b) - \mathbb{P}(o \leftrightarrow b, o \leftrightarrow A)$ is positive; then $\delta \leq 1$, and the last display gives $q\delta \leq (1 - q)\mathbb{P}(o \leftrightarrow b, o \leftrightarrow x) \leq 1 - q$ for every q with $0 < q < 1$. Taking $q = 1 - \delta/4$, which lies between $3/4$ and 1 , turns this into $(1 - \delta/4)\delta \leq \delta/4$, that is $\delta \geq 3$, a contradiction. Hence $\delta \leq 0$, which is (4). $\square$

Remark 9.5. Whether (4), the bound comparing $\mathbb{P}(o \leftrightarrow b, o \leftrightarrow A)$ with $\mathbb{P}(o \leftrightarrow A, a \leftrightarrow b)$, holds for the unique vertex realising the minimum of ρ when A has three or more elements is open. If instead a is chosen to minimise $\mathbb{P}(o \leftrightarrow A, x \leftrightarrow b)$ over $x \in A$, then (4) holds by (1).

10 The Lean Formalization

Every result above is formalized in Lean 4 with Mathlib, on top of the formalization of Theorem 3.3 as `Percolation.Continuity.CSH.cshHolds` in [Dev26]. This section records the correspondence.

Theorem 4.1 is `KNAll.genY_all` and Theorem 5.2 is `KNAll.conjecture4Fixed_holds`.

Corollaries 5.4 and 5.5 are the declarations of `KN/Conjectures.lean` and `KN/ClusterProperty.lean`, among them `conjecture4_holds`, `conjecture2Strong_holds`, `conjecture2_holds`, `conjecture1_holds`, `conjecture3_holds` and `question7_holds`. Question 2.6 is `KNAll.question5_holds`.

Steps 1 to 7 of Section 7.2 are the theorems of `KN/SourceGeneralCSH.lean`, whose conclusion is `KNAll.Guarded.source_guardCSH_nondegenerate`. Step 8 is `KNAll.Guarded.sourceSurplusY_all_disjoint` in `KN/SourceSurplus.lean`, and Step 9 is `KNAll.Guarded.source_fixedMin` in `KN/SourceProjection.lean`; together they give Theorem 7.1 as `KNAll.setSourceFixedMin_holds`. Theorem 7.4 and Conjecture 2.5 are `KNAll.Guarded.conjecture6Strong_holds` and `KNAll.Guarded.conjecture6_holds`. Theorem 8.1 is `KNAll.question9_holds`, obtained from `KNAll.setSourceFixedMin_holds` through `KNAll.question9_of_setSourceFixedMin`. Proposition 9.1 is `KNAll.not_question8EveryMin`. Theorem 9.4 is proved in this document only.

Each declaration compiles under Lean 4.32.0 with Mathlib commit `81a5d257c8e410db227a6665ed08f64fea08e997`, with no `sorry`, and depends only on the axioms `propext`, `Classical.choice` and `Quot.sound`.

All of this was done autonomously by ChatGPT 5.6 Sol and Claude Fable 5.1, with minimal human intervention.

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- [KN24] G. Kozma and S. Nitzan, *A reduction of the percolation at criticality problem to a conjectured inequality*, arXiv:2401.12397, 2024. Conjectures 1 to 3 with display (3) are on pp. 3 and 15, Conjecture 4 and Question 5 on pp. 31–32, Conjecture 6 on p. 34, and Questions 7 to 9 with display (41) on p. 36.